ChronoScope: Sequence Diagnostics and Empirical Forecast Evaluation

Felipe Santibáñez-Leal
CAOS open-research programme, Santiago, Chile
ORCID: 0000-0002-0150-3246

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`https://github.com/fsantibanezleal/CAOS_RES_ChronoScope`.

Abstract

ChronoScope combines source-bound sequence diagnostics with distinct forecast evaluations. Exact Matrix Profile and FLUSS annotations, multiscale entropy, symbolic complexity, time-reversal statistics and seeded nonlinear estimates characterize original observations. A separate M4 Daily study admits 4,226 of 4,227 identities, confirms 21 of 40 prefix features under a fixed two-partition rule, and retains a negative selection result: mean confirmation MASE 3.3501 for the selector versus 3.2570 for a discovery-selected AutoETS constant. The fifteen-case forecasting experiment likewise retains the fixed ContextMemory negative result, mean MASE 1.4504 versus 1.4489 for RecencyMean. Two additional combinations select parameters using causally completed historical validation blocks. In an external twelve-method comparison, ValidationMix has mean MASE 0.2855 versus RecencyMean's 0.3109 on 294 eligible M4 Weekly series, and 0.8993 versus 0.9085 on 1,422 unique M3 Monthly series. AutoTheta remains better on monthly point error and both collections' interval score. Independent recomputation checks all 53,148 external forecasts. A reconstructed one-step interval study separately distinguishes local coverage error from error after temporal averaging. Normative schemas, generated browser types, complete numerical files and sixteen executed bilingual lessons expose computations and limitations. These finite-collection improvements do not establish population superiority, calibrated uncertainty, operational benefit or priority for the underlying combination principles.

1 Decision problem and scope

Inventory, staffing and capacity commitments depend on forecasts made before demand is observed. A forecast revision can alter those commitments even when its statistical benefit has not been measured. The relevant engineering question is therefore broader than selecting the smallest displayed error: which observations were available, what algorithm produced the path, which future outcomes were scored, and what uncertainty remains before an action is taken?

ChronoScope supports inspection of these questions through selected real excerpts, generated mechanisms and explicit controls. Generated cases explain mechanisms and expose defects; they are not customer observations. The real excerpts represent electricity demand, air quality and M4 series. Air quality is a forecasting example rather than a demand-planning measurement. No monetary benefit, service-level improvement, laboratory result or customer deployment is inferred from these examples.

The browser offers series structure, forecasts, interval boundaries, individual errors and historical streaming diagnostics. Documentation separates a sourced executive problem framing from

technical method descriptions. The product is a research and teaching instrument. Acceptance of a production policy would additionally require an operational objective, representative data, governance of revisions, latency constraints and prospective evaluation.

2 Implementation identities

The method roster distinguishes algorithms from their implementations. A reference library, a direct adaptation and a small browser-exported model can share an architectural idea while differing in objectives, preprocessing and training budget. Their results remain separate.

Methods	Implementation and settings	Evidence in the archived comparison
SeasonalNaive, SES, Holt, HoltWinters, Theta	NumPy core; browser TypeScript parity; finite smoothing grids	Recomputed on original contiguous prefixes. Gaussian residual intervals are approximations.
AutoETS, AutoTheta, AutoARIMA	statsforecast; ARIMA stepwise and approximate, context cap 360	Released paths rescored. Separate worked lessons freshly execute the installed adapters.
LightGBM	MLForecast lag table; 100 trees, 31 leaves, learning rate 0.05, context cap 500	Released point-regression path rescored; Gaussian residual intervals. A lesson freshly executes the adapter.
NLinear, DLinear, NHITS	Direct-torch quantile-head adaptations	Released paths rescored; not asserted to be exact replications of the papers.
NLinear (nf), DLinear (nf), NHITS (nf)	neuralforecast reference implementations; fixed 300-step training budget	Released paths rescored; the budget is part of the experiment.
Chronos-Bolt, Chronos-2, TimesFM 2.5, TiRex-2	Pretrained checkpoint adapters; TiRex bridge uses WSL/CUDA	Released paths rescored, without current-checkpoint or pretraining-independence claims.
ContextMemory, RecencyMean, EqualMean	Same five light experts; fixed causal memory and two ablations	Newly computed with matching original prefixes and final targets.

An additional browser NLinear model is exported to ONNX after least-squares fitting on generated windows. It predicts twelve observations from a forty-eight-observation context. It is an independently identified live demonstration, not an additional member of the twenty-two-variant aggregate experiment. On real cases its first twelve held-out targets can be inspected without adapting the fit to those targets. Its interval scale is inherited from its training metadata and does not establish calibration on the real excerpts.

2.1 Corrected lightweight methods

SeasonalNaive repeats the latest observed cycle. SES updates a level, Holt adds a trend, and additive HoltWinters additionally updates a seasonal component. At a prefix length n, the seasonal slot for zero-based future lead k must continue as $(n + k) \bmod m$. Restarting at $k \bmod m$ is incorrect when n is not a multiple of m. An adversarial partial-season example tests this condition independently in Python; the browser parity fixture includes a non-multiple history and a horizon longer than one cycle.

For the nonseasonal Theta implementation, the corrected point equation is

$$\widehat{y}_{n+h} = \ell_n + \frac{b}{2}\left[h - 1 + \frac{1-(1-\alpha)^n}{\alpha}\right], \qquad h \geq 1, \tag{1}$$

where ℓ_n is the selected SES level and b is the fitted linear slope. An independent two-Theta-line construction verifies the finite-history term following Hyndman and Billah [1]. This local method does not inherit AutoTheta's automatic seasonal adjustment merely because the names are related.

The Gaussian widening rule maps a residual standard deviation to three ordered columns at probabilities 0.1, 0.5 and 0.9. Residual estimation, Gaussian shape and horizon widening are assumptions. Neither an ordered matrix nor a quantile-loss fit establishes marginal or conditional calibration. These distinctions also apply to the direct neural adaptations and the lag-tree adapter.

3 Diagnostic atlas and computational boundaries

The diagnostic toolkit contains ten families: stationarity, autocorrelation, seasonality, filtering and decomposition, change points, volatility, distribution and complexity, fractals, nonlinear dynamics, and causality. Methods include ADF/KPSS and related tests, ACF/PACF and Ljung–Box diagnostics, STL/MSTL and periodograms, offline change-point procedures, ARCH/GARCH diagnostics and surrogate-based nonlinear analyses. A short series or unavailable dependency produces an explicit unavailable result for that analysis; it does not silently produce a substitute measurement.

Statistical diagnostics are descriptive or model-dependent. An autocorrelation band containing the observed coefficients does not prove white noise. A unit-root test does not certify future stationarity. A fractal estimate is sensitive to sample length, detrending and the assumed stochastic class. A nonlinear surrogate rejection does not uniquely identify deterministic chaos. The workbench exposes these cautions alongside the results rather than presenting a diagnostic label as a forecasting guarantee.

The browser recomputes lightweight diagnostics and light forecasts for generated scenarios. Committed analysis artifacts retain their original scope; several describe the complete observed series, including what later becomes a forecasting holdout. They are suitable for retrospective explanation, not for feature selection at the earlier forecast origin. The proposed combination computes its own prefix-only contexts and does not consume those retrospective diagnostics.

Heavy statistical, neural and pretrained calculations run offline. GPU availability, checkpoint caches and Python environments are distinct from the served browser bundle. A local environment with only NumPy cannot be represented as a complete neural or foundation-model rerun. Conversely, rescoring a stored model path does not require pretending that its original runtime has been restored.

4 Data provenance and temporal reconstruction

The corpus contains eleven generated cases and four real excerpts. Generated mechanisms cover seasonality, trend, intermittent arrivals, a level shift, multiple seasonalities, heteroscedasticity, long memory, a delayed nonlinear system and a known-future exogenous driver. White noise and a near-random walk remain visible as controls. The exogenous-driver comparison requires a declared arrival policy; univariate forecasts do not gain access to future covariates automatically.

Table 2: Admitted cases: original contiguous history length, horizon and seasonal lag.

Case	History	Horizon	Lag	Source
BRKV_level_shift	288	12	12	Generated
CHAO_mackey	488	12	1	Generated
CTRL_white_noise	148	12	1	Generated
EXOG_promo	226	14	7	Generated
HETV_garch	488	12	1	Generated
INTM_demand	188	12	1	Generated
LMEM_fractional	488	12	1	Generated
MSEA_daily_weekly	648	24	24	Generated
REAL_electricity	456	24	24	Real excerpt
REAL_m4_daily	386	14	7	Real excerpt
REAL_m4_hourly	456	24	24	Real excerpt
REAL_pm25	456	24	24	Real excerpt
RWLK_noise	188	12	1	Generated
SEAS_hourly	456	24	24	Generated
TRND_seasonal	168	12	12	Generated

The source manifest records provenance, redistribution permission, seed, original series descriptors, artifact paths and byte counts. Real-data CSVs are committed as compact admitted excerpts. A source's redistribution status is an explicit field; a restricted source must not expose its raw path merely because aggregate metrics are publishable. The current comparison uses only available, non-redacted inputs.

4.1 Why a replay history is not a fitting series

The released trace stores at most three hundred historical display samples and their original indices. Nine case histories exceed that display limit. Treating those samples as consecutive observations changes the time interval and can destroy the meaning of a seasonal lag. It also changes a historical scale used by MASE. The initial comparison made this error and is invalidated; its earlier bytes remain identifiable through Git, not as admitted scientific evidence.

The corrected protocol reconstructs the original series from each committed CSV or deterministic case definition and its manifest seed. At every displayed index the reconstructed observation must agree with the released rounded value within 0.000051. The final targets must also agree at that tolerance, and the original prefix length must match the trace's declared history length. All fifteen reconstructions satisfy these checks. The complete contiguous prefix is used for light fitting and historical scaling. Original per-engine context caps remain explicit; a common available prefix does not mean every algorithm consumes the same number of observations.

The reconstructed history retains source precision. Frozen targets and model paths retain their released four-decimal precision. This small rounding distinction is disclosed and is not the much larger temporal-decimation error. Source and implementation hashes bind the corrected artifact to the exact inputs. A replay-only array whose declared length exceeds its display length is refused as a consecutive fit input.

5 Common final-block protocol

Each selected series supplies one forecast origin and one final block. Fourteen heavy forecast paths remain frozen; five corrected lightweight methods and three combinations are newly computed.

Every admitted path is scored on the same actual values and horizon within its case. Original rolling-backtest results remain archived. Their differing origin budgets, ranging from three to more than twenty-five windows, make a direct ranking of those aggregates confounded by the evaluation schedule.

Let $y_1, \ldots, y_n$ denote the contiguous prefix, $a_1, \ldots, a_H$ the final targets and p_j a point prediction. The historical seasonal scale is

$$s_m = \frac{1}{n-m} \sum_{t=m+1}^{n} |y_t - y_{t-m}|, \tag{2}$$

with the lag bounded to the available history. MAE is $H^{-1} \sum_j |a_j - p_j|$, RMSE is the square root of mean squared error, and MASE is MAE/s_m [2]. A value below one compares a future error with a historical scale. It does not directly show that the method beat an independently scored SeasonalNaive path on the same future block. A zero scale produces an unavailable scaled metric rather than an invented denominator.

For lower and upper columns l_j, u_j, empirical coverage is $H^{-1} \sum_j \mathbf{1}\{l_j \le a_j \le u_j\}$ and width is $H^{-1} \sum_j (u_j - l_j)$. With nominal interval miscoverage $\alpha = 0.2$, the interval score is

$$IS_j = (u_j - l_j) + \frac{2}{\alpha}(l_j - a_j)_+ + \frac{2}{\alpha}(a_j - u_j)_+. \tag{3}$$

MSIS is its block mean divided by s_m. The artifact's WQL uses only $Q = \{0.1, 0.5, 0.9\}$:

$$\mathrm{WQL}_3 = \frac{2 \sum_{q \in Q} \sum_{j=1}^{H} \rho_q(a_j - \widehat{y}_j^{(q)})}{3 \sum_{j=1}^{H} |a_j|}, \quad \rho_q(e) = \max(qe, (q-1)e). \tag{4}$$

This is an average of three quantile losses, not a complete predictive-density evaluation. A zero absolute-target sum makes this normalized score unavailable. Proper-score theory motivates reporting probabilistic quality, while the finite quantile grid limits what is observed [3].

6 A fixed context-memory proposal

ContextMemory tests whether similar completed historical contexts help estimate expert competence beyond recency alone. Feature-dependent combinations and dynamic ensembles are established ideas [4, 5]. The experiment tests a specific within-series recurrence rule and makes no priority claim for contextual weighting.

The expert set is fixed to SeasonalNaive, SES, Holt, HoltWinters and Theta. Up to twelve completed historical origins are selected within the prefix. For every origin o, eligibility requires $o + H \le n$. A normalized half-window mean contrast, level shift, volatility-change statistic and seasonal correlation describe the context. Expert multi-step absolute losses are normalized using information available at that origin. A Gaussian context-distance kernel combines with recency weights of half-life six origins. Local estimated losses are shrunk 25% toward the recency-only loss estimate.

An exponential loss transformation produces a simplex allocation, followed by a 5% uniform floor. With $K = 5$ experts, each final weight is at least $0.05/K = 0.01$ and the weights sum to one. Corresponding point and quantile columns are averaged with these same weights. Nonnegative convex weights preserve the order of the input quantile columns. They do not make the result a calibrated predictive mixture.

RecencyMean removes context similarity while retaining the experts, historical origins, recency weighting and competence transformation. EqualMean assigns a uniform allocation. These matching ablations distinguish the contextual component from the more general benefit of

combining methods. A positive scale and location transformation supplies a structural check away from numerical degeneracy thresholds. Completed-target eligibility and quantile ordering have independent adversarial tests.

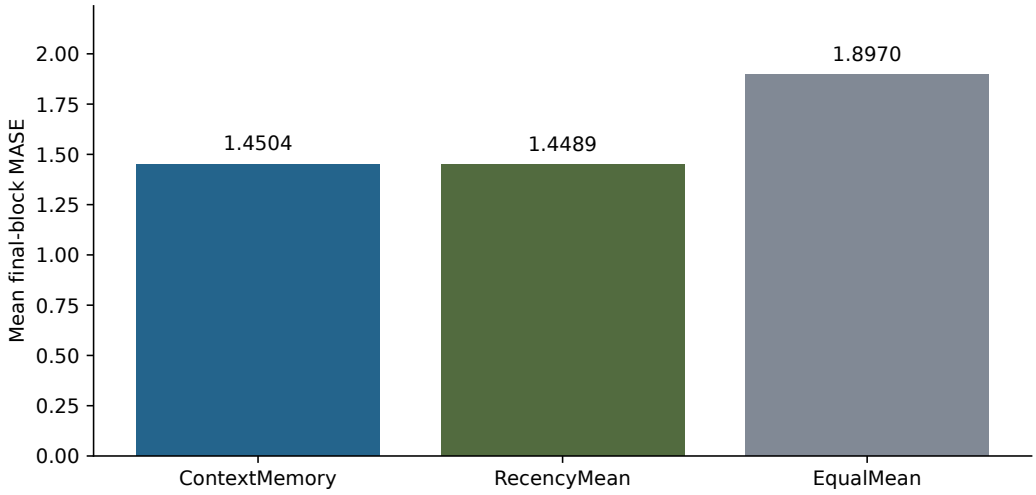


Figure 1: Fixed matching-expert combinations on the corrected fifteen-case protocol. Lower mean MASE is preferable. The contextual component does not improve on the recency-only ablation. Bars summarize selected final blocks, not independent population replications.

7 Archived-path final-block results

Table 3: Corrected common final-block comparison. Means weight each selected case equally.

Method	Cases	Mean MASE	Mean coverage
Chronos-Bolt	15	1.0363	0.7563
TimesFM-2.5	15	1.0633	0.8071
TiRex-2	15	1.0809	0.7694
Chronos-2	15	1.1555	0.6905
DLinear	15	1.1600	0.5595
AutoARIMA	15	1.2104	0.7813
NLinear	15	1.2298	0.4524
NHITS (nf)	15	1.2546	0.5079
NHITS	15	1.2898	0.4857
LightGBM	15	1.2983	0.5786
SeasonalNaive	15	1.4137	0.8702
RecencyMean	15	1.4489	0.8889
ContextMemory	15	1.4504	0.8944
AutoTheta	15	1.4964	0.7329
HoltWinters	15	1.6778	0.8639
AutoETS	15	1.6940	0.7639
NLinear (nf)	15	1.7547	0.1821
EqualMean	15	1.8970	0.8611
SES	15	1.9167	0.8274
Theta	15	1.9224	0.8274
Holt	15	3.9846	0.7302

Table 3: Corrected common final-block comparison. Means weight each selected case equally.

Method	Cases	Mean MASE	Mean coverage
DLinear (nf)	15	10.7385	0.4996

Chronos-Bolt has the smallest mean MASE in this selected, mixed-provenance comparison, followed by TimesFM 2.5 and TiRex-2. Their paths were not freshly refitted or regenerated for this experiment. No statistical significance, resource-normalized superiority or independence from pretraining data follows from that order. Method-specific training budgets and context limits remain part of the compared configurations.

Coverage exposes substantial differences that a point ranking conceals. Several neural paths have narrow intervals and low observed coverage. A broad interval can increase coverage at the expense of decision usefulness; coverage alone should not be optimized. Individual case outcomes and widths remain available in the artifact and workbench, including unfavorable neural and combination results.

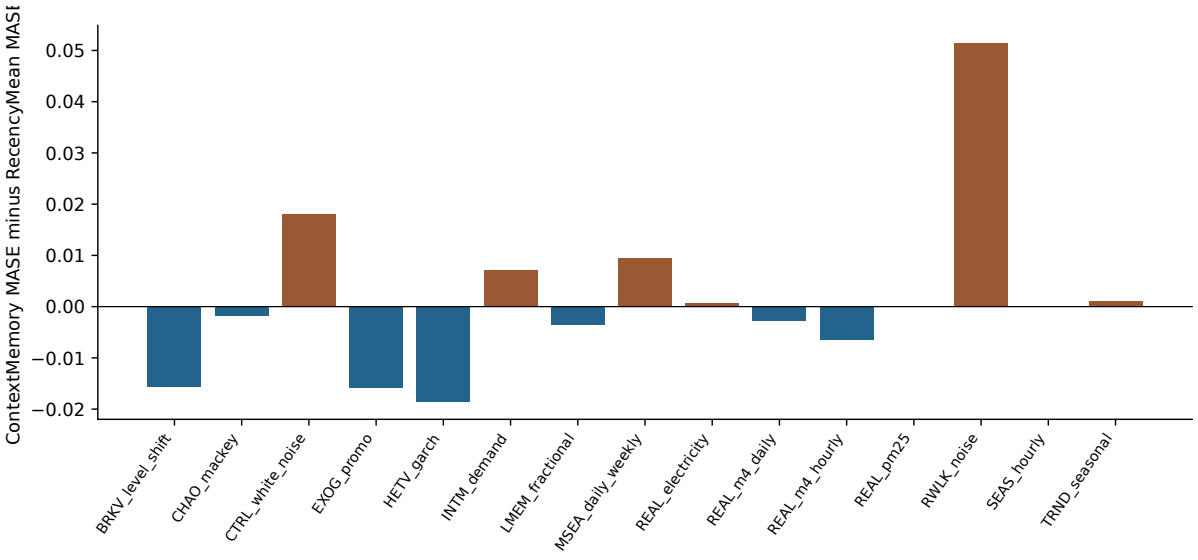


Figure 2: ContextMemory minus RecencyMean MASE for every selected case. Positive values favor the recency-only ablation. The common target and expert set isolate the contextual weighting component within each case, but the selected origins do not establish a general effect.

ContextMemory's mean MASE is 1.4504, versus 1.4489 for RecencyMean. The difference is unfavorable at the declared primary criterion, despite occasional individual improvements. EqualMean gives 1.8970. Settings are unchanged after either the invalidated display-history calculation or the corrected run. A changed configuration would define a new experiment and must not overwrite this result.

8 Fresh execution of the complete roster

A second experiment fits or infers all twenty-two configurations again from the same verified contiguous prefixes. The separately admitted artifact records every prediction, source-file and input hash, checkpoint revision, package version, fit/inference duration and warning category. An independent standard-library checker recomputes all seven metrics from the stored numerical arrays, verifies complete inventories and rejects trials, changed targets, frozen paths disguised as

new fits and stale source bindings. These admission checks establish internal numerical consistency and recorded identity; they do not establish independence from model pretraining data.

The execution uses Python 3.12, NumPy 2.1.3, PyTorch 2.9.1 with CUDA 12.8, StatsForecast 2.0.3, LightGBM 4.6.0, MLForecast 1.0.31, NeuralForecast 3.1.9, Chronos forecasting 2.3.2, TimesFM 2.0.2 and TiRex-2 0.1.1. Its GPU is an NVIDIA GeForce RTX 4070 Laptop GPU. NumPy follows the TiRex-compatible line in this isolated Linux environment; notebook/parity computations use their separately pinned environment. Direct and reference neural models retain their declared 300-step budgets. LightGBM uses one worker; BLAS/OpenMP threads and compiler workers are bounded, and torch CPU operations use four threads. These resource choices are part of the recorded configuration.

Table 4: Fresh execution alongside the archived-path comparison. The same fifteen final blocks are used in both; these are not independent replications.

Method	Archived MASE	Fresh MASE	Fresh coverage
Chronos-Bolt	1.0363	1.0363	0.7563
TimesFM-2.5	1.0633	1.0633	0.8071
TiRex-2	1.0809	1.0809	0.7694
Chronos-2	1.1555	1.1555	0.6905
DLinear	1.1600	1.1600	0.5595
AutoARIMA	1.2104	1.2104	0.7813
NLinear	1.2298	1.2298	0.4524
LightGBM	1.2983	1.2983	0.5786
NHITS (nf)	1.2546	1.2993	0.4968
NHITS	1.2898	1.3031	0.4746
SeasonalNaive	1.4137	1.4137	0.8702
RecencyMean	1.4489	1.4489	0.8889
ContextMemory	1.4504	1.4504	0.8944
AutoTheta	1.4964	1.4964	0.7329
HoltWinters	1.6778	1.6778	0.8639
AutoETS	1.6940	1.6940	0.7639
NLinear (nf)	1.7547	1.7547	0.1821
EqualMean	1.8970	1.8970	0.8611
SES	1.9167	1.9167	0.8274
Theta	1.9224	1.9224	0.8274
Holt	3.9846	3.9846	0.7302
DLinear (nf)	10.7385	10.7385	0.4996

The two experiments use the same final targets, so rerunning an implementation is not another independent statistical test. Differences can reflect runtime, precision or training implementation rather than a changed data population. ContextMemory still fails its aggregate comparison with RecencyMean; the selected settings are not retuned after that outcome. Warning categories remain visible even when predictions are finite. Durations include fitting, inference, imports and first-use loading and cannot be interpreted as a normalized warm-inference leaderboard.

Exact Hugging Face revisions identify the four admitted pretrained checkpoints. TimesFM 3.0 is a newer upstream candidate with noncommercial, nonproduction weight terms; the admitted experiment retains the Apache-licensed 2.5 checkpoint. This roster therefore makes no claim to contain the latest released model of every family. The original streaming results below were not regenerated by this final-block execution.

9 Combinations selected by temporal validation

ContextMemoryCV and ValidationMix extend the fixed combination with parameter selection inside the observed prefix. They retain SeasonalNaive, SES, Holt, additive HoltWinters and finite-history Theta as experts. At a remembered validation origin v, a training block beginning at o is eligible only if $o + H \leq v$. At least three fully completed training blocks are required. Candidate scores average scaled absolute error over these causally held-out historical blocks. Without eligible validation, the method falls back to the original RecencyMean rule. No outer target is accepted by the fitting API.

ContextMemoryCV selects among context shrinkages $\{0, .5, 1\}$ and temperatures $\{.15, .5, 1\}$ plus the recency-only and equal-weight controls. ValidationMix additionally considers regularized convex stacking, minimizing mean squared normalized expert-combination error plus $\lambda\|w - \mathbf{1}/5\|_2^2$ over the simplex, for $\lambda \in \{.01, .1, 1, 10\}$. Five experts permit explicit active-set enumeration. A 5% uniform floor is applied after fitting. Candidate ranking uses historical absolute error; the selected rule is refit on all completed memory blocks. All families and parameter values are fixed before the external outcomes are evaluated. Established stacking and temporal-validation principles are not claimed as new general ideas.

The external sources are complete Monash M4 Weekly and M3 Monthly inventories [20, 21]. Horizons and seasonal periods are $(13, 52)$ and $(18, 12)$, respectively. Every method receives at most the final 512 contiguous observations at an origin. Up to three nonoverlapping final blocks are used, with a minimum available prefix of $\max(2m, 2H + 8, 24)$. Sixty-five weekly identities have no eligible origin, leaving 294 series and 882 origins. Six exact duplicate monthly identities are excluded, leaving 1,422 series and 3,547 origins. Every source identity and exclusion remains available. Origins are averaged within series, then series equally within each collection. This is not the original forecasting-competition protocol or an official competition ranking.

Table 5: Complete external comparison. W denotes 294 eligible M4 Weekly series; M denotes 1,422 unique M3 Monthly series. MASE and MSIS average origins within series and then series equally; Cov is empirical interval coverage. These are descriptive summaries, not confidence intervals.

Method	W MASE	W MSIS	W Cov	M MASE	M MSIS	M Cov
SeasonalNaive	0.8446	8.1670	0.9326	1.0864	8.8123	0.9342
SES	0.3533	2.3404	0.8926	1.0456	7.1945	0.8534
Holt	0.3806	2.4611	0.8844	1.8040	10.8472	0.8005
HoltWinters	0.5396	3.0778	0.8413	0.9375	6.7112	0.9160
Theta	0.3469	2.3237	0.9041	1.0021	7.0453	0.8700
AutoETS	0.3413	1.8490	0.8713	0.8514	4.1518	0.7324
AutoTheta	0.3076	1.7125	0.8511	0.8472	4.2994	0.7279
ContextMemory	0.3079	2.9468	0.9751	0.9086	6.8846	0.9348
RecencyMean	0.3109	2.9315	0.9742	0.9085	6.8846	0.9348
EqualMean	0.3548	3.3233	0.9702	0.9962	7.2131	0.9333
ContextMemoryCV	0.2848	2.5538	0.9511	0.8996	6.7999	0.9329
ValidationMix	0.2855	2.5477	0.9503	0.8993	6.8208	0.9326

ValidationMix lowers mean MASE relative to RecencyMean on both admitted collections, .2855 versus .3109 weekly and .8993 versus .9085 monthly. ContextMemoryCV gives .2848 and .8996. AutoTheta's .3076 weekly result is higher, while its .8472 monthly result is lower. The point improvement therefore depends on collection and comparator. Context shrinkage, temperature and temporal selection change together; this comparison does not isolate the causal value of the context kernel alone.

Interval quality remains unfavorable relative to AutoTheta. ValidationMix gives weekly MSIS 2.5477 and coverage .9503, and monthly MSIS 6.8208 and coverage .9326. AutoTheta gives 1.7125

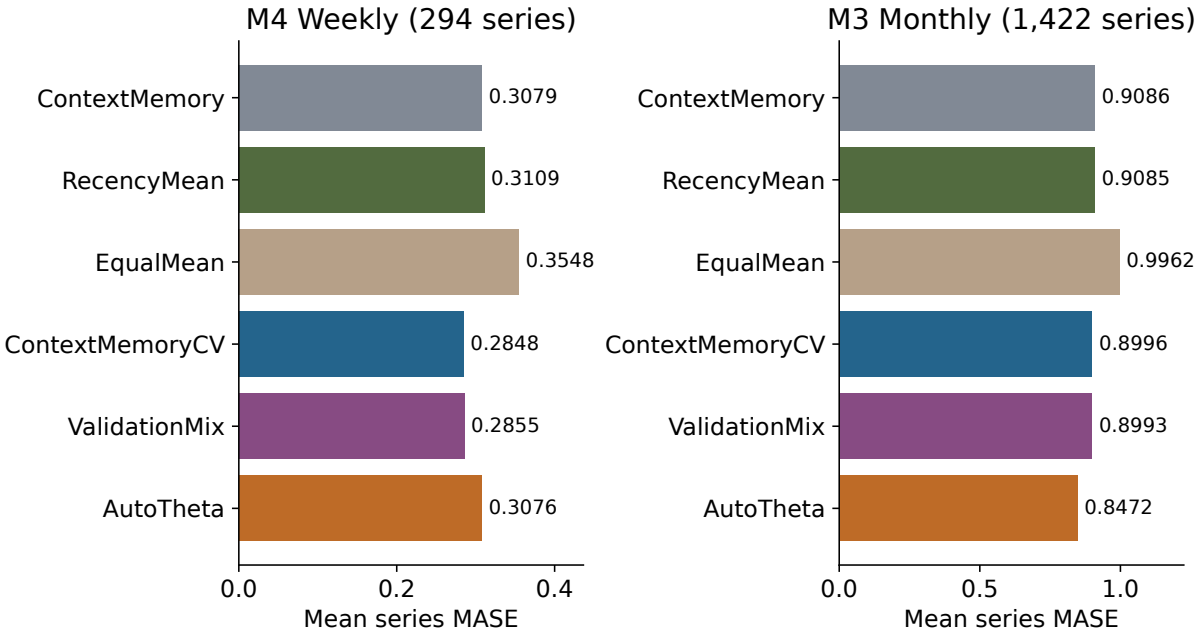


Figure 3: The new temporally selected combinations and fixed controls on each external collection, with AutoTheta as an automatic-statistical comparator. The complete twelve-method table retains the other individual experts and AutoETS. Bars are descriptive mean MASE without population confidence intervals.

and .8511 weekly, and 4.2994 and .7279 monthly. Nominal coverage is .8. These weighted Gaussian endpoints are not calibrated by minimizing point error, and wider intervals do not establish a better decision rule.

On the original fifteen development blocks, ContextMemoryCV and ValidationMix give mean MASE 1.3974 and 1.4031, while the original fixed ContextMemory and RecencyMean results remain 1.4504 and 1.4489. Their MSIS values, 9.5025 and 9.3428, are worse than RecencyMean's 9.1076. These already examined cases are not new external confirmation. The separate original twenty-two-variant execution and complete M4 Daily association/selection study remain unchanged.

All 53,148 external forecast blocks are independently rescored against their source-derived targets and seasonal scales. Final expert weights reconstruct the combined quantile paths. Temporal splits, complete historical targets, candidate losses and chosen parameters remain in the compressed numerical file. Browser delivery uses a compact index and separately fetched per-series documents, rather than downloading the entire numerical file for one chart. Every displayed path is independently checked against its stored score. Exact duplicates do not imply that all remaining series are independent or free of related-source overlap; no population or operational-superiority conclusion follows.

10 Streaming evaluation as separate evidence

The historical streaming experiment uses a predict–observe–update order [6], but the point forecaster is recomputed on each full observed prefix. Its artifact key `Theta` denotes a smoothing-plus-half-drift adaptation, not the corrected finite-history Theta equation used in the final-block comparison. ACI-style and PID-style controllers adapt the effective miscoverage level of a normalized-residual FIFO buffer. The PID-style rule is not a faithful implementation of published Conformal PID [8]; its guarantees are not imported by name. No native stateful model update or framework capability comparison is evaluated.

A full numerical reconstruction exports 21,296 method-cutoff rows across fifteen cases, with targets, forecast endpoints, scores, buffer sizes and controller states. Every archived coverage trajectory and final weighted quantile loss matches at saved precision. At nominal 80%, mean rolling absolute coverage error is .0830 for the raw smoothing-plus-drift base, .0656 for its ACI-style controller and .0645 for its PID-style controller. The much smaller .0416, .0057 and .0060 values measure absolute deviations after temporal averaging, which permits local errors to cancel. These are different statistics. Scaled widths and interval scores accompany coverage.

The reconstruction also varies coverage (.7, .8, .9), two point bases and three gain multipliers, retaining all 630 case-setting rows. Eleven generated mechanisms are repeated at two additional seeds, while the four real excerpts are not assigned artificial repetitions. These are sensitivity calculations on the same selected design, not independent population confirmation. The original gains have no recovered historical tuning rationale, and original per-call hardware metadata remain unavailable. Historical latencies therefore support no normalized runtime comparison. Exact source arrays and hashes permit numerical replay without claiming to restore missing acquisition history. These streaming outcomes are separate from both the final-block experiment and the M4 Daily corpus study.

11 Artifact and browser engineering

Twelve catalog boundaries cover traces, manifests, the index, diagnostic analysis, streaming results, ONNX metadata/assets, archived-path and fresh comparisons, historical selector metadata, the complete daily corpus study, the validated-combination index and its per-series records. Normative JSON Schemas validate the committed JSON documents and generate browser TypeScript types through an OpenAPI component inventory. That inventory supports type generation without activating a prediction API. Cross-field invariants, source hashes, temporal reconstruction and numerical parity remain executable tests; a structural schema alone cannot enforce them.

The browser delegates chart interaction, summaries, tables, reset controls and keyboard support to the pinned shared shell. The product supplies data adapters, stable method identities and scenario state. Bilingual content uses locale-aware numbers; model and licence versions retain their identity. Request cancellation and generation identifiers prevent a slower case request or inference result from overwriting a newer selection.

Numbered local workflows check prerequisites, initialize declared environments, build an external data candidate and start the development server. Generation binds source and payload fingerprints, refuses partial inventories and changed frozen inputs, and separates candidate creation from explicit promotion. Historical engine-version metadata is retained rather than relabeled to match the new browser release. An unused epidemic-model surrogate inherited from an older repository structure was removed.

Sixteen executed English and Spanish notebooks cover a seasonal reference, an incomplete-season experiment, actual statistical/tree adapter execution, the fixed combination with its ablations, sequence patterns, multiscale complexity and disjoint-series corpus inference. Their metadata records exact direct dependencies, sources, inputs and method coverage. Fresh-kernel execution and byte comparison check that committed outputs are current. Numerical convergence notices are retained as portable text; machine paths and incidental notebook metadata are excluded.

12 Validation and remaining scientific limits

The validation layers are distinct. Structural schema checks establish artifact shape; source bindings establish input identity; independent equation tests establish selected numerical behaviours;

browser traversal measures rendered routes and interactions; local notebook reproduction measures executable teaching artifacts. A successful build alone establishes none of the other layers. Computational records distinguish local numerical checks from remote CI, deployment and owner acceptance.

The fifteen-case forecasting experiment has these scientific limitations: one origin per selected series; a small, non-random collection; configuration-specific runtime and historical evidence retained in a separate experiment; unverified pretrained-data overlap; bounded historical training budgets; approximate lightweight uncertainty; possible numerical convergence issues in freshly executed reference adapters; and retrospective diagnostic artifacts that may include the final block. No operational acceptance or economic benefit is implied. For those twenty-two configurations, a broader study still needs predeclared datasets, repeated origins, runtime and memory budgets, dependence-aware comparisons and independent pretraining-overlap checks. The M4 Daily study below extends series-level evidence for its separate eight-method roster.

For the contextual proposal, a stronger follow-up also needs faithful FFORMA and arbitrated-ensemble comparators rather than only internal ablations. Such work could reject or improve the proposed recurrence rule. The current negative outcome remains informative without being promoted to a general impossibility result.

13 Reproduction and availability

Implementations and admitted numerical outputs are available in the frozen source and data snapshot.

The source tree contains the corrected computation in `chronoscopelab.reevaluation`, the fixed proposal in `model/context_memory.py`, original released traces, admitted source CSVs and the generated comparison. Running `scripts/reevaluate.py` regenerates the common final-block artifact while preserving the original traces. The staged local workflow provides complete candidate admission and explicit promotion. Exact commands, environment pins and validation results are recorded with the source.

The numerical tables and figures are generated directly from the two admitted forecast comparisons, full-sequence diagnostics and complete M4 study. The research report generator checks independent corpus statistics before producing its tables. The new lesson authoring script records the exact admitted corpus results; fresh-kernel execution remains a separate step. Code and compact admitted data use the repository's software/data licences; prose uses the licence stated in the header. Numerical reproduction does not establish representative population sampling, operational adoption or calibrated uncertainty for a future deployment.

14 Sequence diagnostics on original observations

The diagnostic extension operates on every original observation rather than the display-decimated history. For window length m, the exact self-join Matrix Profile is

$$P_i = \min_{j:\,|i-j|>\lceil m/4\rceil} \|z(x_{i:i+m}) - z(x_{j:j+m})\|_2, \tag{5}$$

where each window receives its own population-standard-deviation normalization. STUMPY supplies the exact join and the constant-subsequence convention. A completely constant sequence is reported as unavailable because it cannot discriminate local shapes. The motif is the lowest-distance reported nearest-neighbor pair whose two windows do not overlap. The three discord candidates have large finite profile distances and non-overlapping query windows. Their ranks

are not calibrated anomaly probabilities; normalization deliberately removes local amplitude and mean [14].

FLUSS consumes nearest-neighbor indices and constructs a corrected arc curve [15]. The case diagnostics request three candidate regimes with exclusion factor one and $L = m$. No known generator boundary enters this calculation. The requested regime count is a parameter, not an estimated count of true changes. App overlays retain original integer window positions, inclusive interval edges and candidate boundaries. These retrospective annotations use the complete sequence and cannot establish an online detection time.

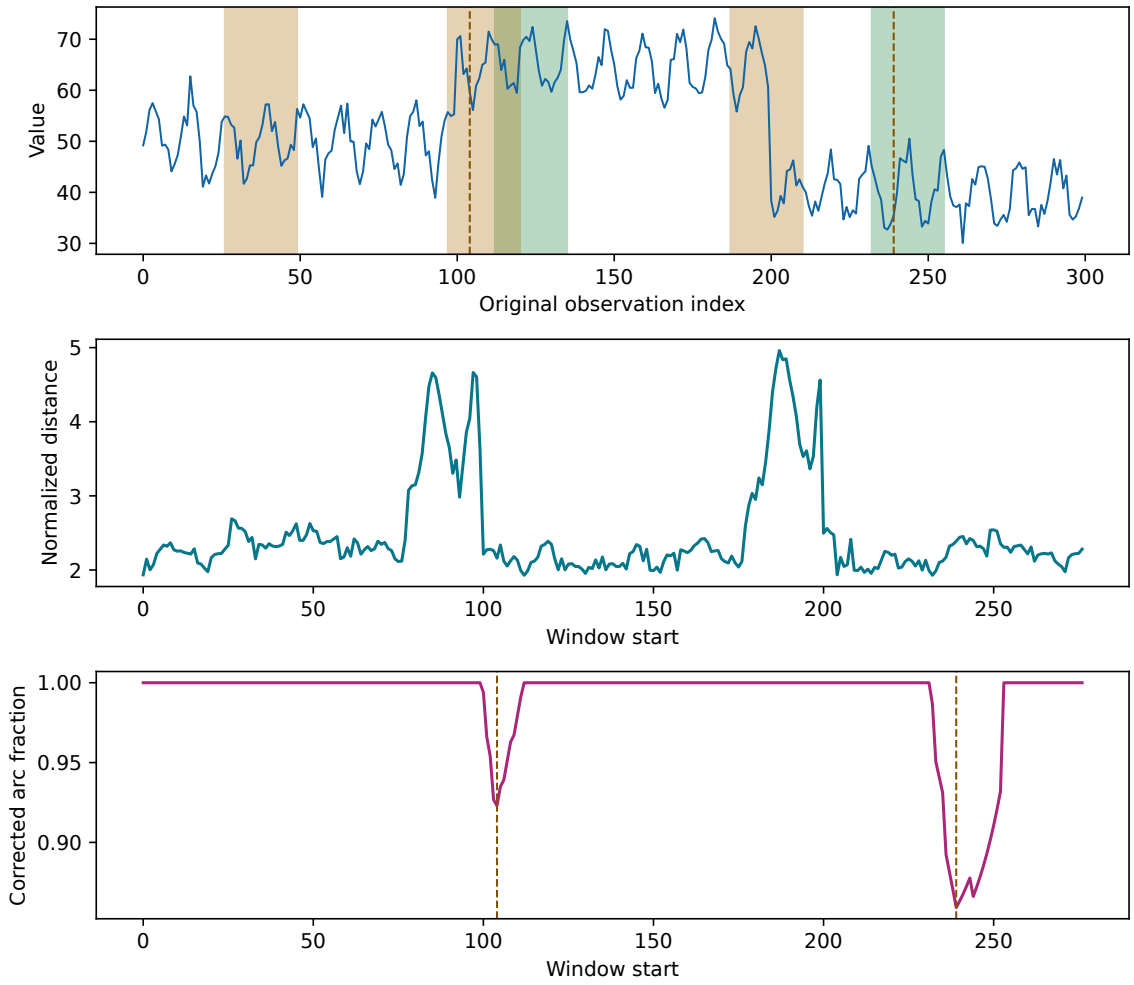


Figure 4: Original-axis sequence annotations for the structural-break case. Colored intervals identify a motif pair and discord candidates; dashed boundaries are FLUSS candidates. Neither the window ranks nor the requested segmentation count establishes ground-truth anomalies.

Costa multiscale entropy averages non-overlapping blocks at each scale τ [16]. Sample entropy uses order two, tolerance $r = 0.15s(x)$ fixed to the original population standard deviation, and the same $N_\tau - 2$ template starts for dimensions two and three. Distinct unordered pairs match at Chebyshev distance at most r, excluding self matches. If B_τ pairs match at dimension two and A_τ extensions match, the estimate is

$$\mathrm{SampEn}_\tau = -\log(A_\tau / B_\tau). \tag{6}$$

Fewer than twenty coarse observations, or no matching extensions, yield an unavailable estimate with a reason and exported counts. An unavailable value is not zero complexity. Constant sequences give zero where all eligible templates match.

Binary Lempel–Ziv complexity uses $1[x_t > \text{median}(x)]$, assigning ties to zero. Both phrase count $c(n)$ and $c(n)\log_2(n)/n$ are retained [17]. The asymptotic normalization is not clipped to one. Time-reversal asymmetry reports the mean of $x_{t+\ell}(x_{t+2\ell}^2 - x_t^2)$ at lags one, two and seven, where available, on raw and standardized observations. Raw units are cubed source units. Reversal changes the sign, but a finite nonzero statistic alone is not a significance test or an irreversibility test.

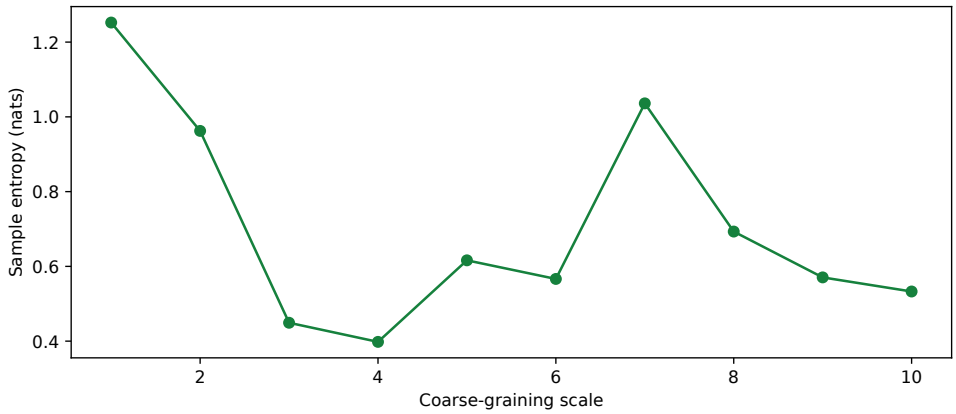


Figure 5: Multiscale entropy on the structural-break case with the original tolerance fixed across scales. Coarse sequence lengths decrease with scale; finite-sample availability and matching counts remain part of the interpretation.

The previous nolds correlation-dimension and Lyapunov defaults used randomized RANSAC fits without an explicit random state. The adapter now obtains deterministic log-distance fitting data, then fits the same robust polynomial-feature linear model with an explicit local seed. Least squares is the fallback when consensus fails. A case seed is derived from SHA-256 of case identity and the supplied pipeline seed; surrogate and 0–1 calculations receive derived local seeds. Process-global random state is not changed. DFA and fallback R/S slopes explicitly request least squares. These fit choices can move historical diagnostics even when the source observations are unchanged. Two complete analyzer calls for every case must serialize identically before promotion; independent-process and caller-RNG tests additionally protect the estimator boundary. The existing surrogate gate remains necessary for a chaos classification.

15 Corpus forecastability and disjoint-series confirmation

A separate fixed-protocol experiment processes the complete Monash M4 Daily source, 4,227 series under CC BY 4.0 [18]. Every identity is admitted or assigned an explicit exclusion. Exact whole-sequence duplicates retain the lexicographically first identity for inference. Nonfinite observations, insufficient prefixes and unavailable MASE denominators are excluded with reasons. Source ZIP and extracted TSF hashes identify the input bytes; a full-sequence identity hash is used only for duplicate grouping and partition allocation, never as a predictor.

Each admitted series contributes three non-overlapping fourteen-step future blocks. Each model uses at most the preceding 1,024 contiguous observations at an origin. Features use only the earliest fitting prefix. Forecast seasonality is one, matching the M4 Daily setting; weekly phase strength is a separately named diagnostic. The fixed roster is Naive, SeasonalNaive, SES, Holt, HoltWinters, Theta, AutoETS and AutoTheta. Some configurations coincide when the period is one; these explicit identities do not imply eight distinct future paths. No deep or pretrained model participates in this corpus study, which does not replace the separate twenty-two-variant case experiment.

Model MASE is averaged over the three origins within each series. The association target is $\log(1 + \min_k \overline{\mathrm{MASE}}_k)$ over the fixed roster. This is an optimistic within-roster oracle, not inherent predictability. Forty prefix diagnostics include catch24, location/scale and ordering summaries, multiscale entropy, binary complexity, reversal asymmetry and profile distances. Associations may reflect a scaled-error denominator or source grouping rather than a causal forecasting mechanism.

Exact sequence identities allocate approximately seventy percent of series to discovery and the rest to confirmation. Kendall tests use available feature values and aggregate series targets. Benjamini–Yekutieli correction treats the entire feature family separately in each partition at $q = 0.05$ [19]. In sorted order, adjusted probabilities use

$$q_{(i)} = \min_{j \geq i} \left\{ 1, \frac{K H_K}{j} p_{(j)} \right\}, \qquad H_K = \sum_{k=1}^{K} k^{-1}. \tag{7}$$

Confirmation requires rejection in both partitions and matching effect direction. All feature counts, signed effects, raw probabilities, adjusted probabilities and negative decisions are retained. Series are the statistical units; origins are not treated as independent replications. Related industries or latent sources can still violate observation-independence assumptions after exact deduplication.

A fixed random forest predicts the best model identity from discovery features, with 200 trees, maximum depth eight, minimum leaf size ten and random state 42. Missing-value medians and the constant comparator are fitted on discovery only. Confirmation outcomes choose neither parameters nor feature thresholds. The selected rule is compared once on confirmation series with that constant method and the future-informed per-series oracle. Scores are descriptive held-out results, not population-superiority or operational guarantees. Negative selection outcomes remain in the report.

The complete source contains 4,227 identities; 4,226 are admitted and 1 is explicitly excluded. The deterministic partitions contain 2,946 discovery and 1,280 confirmation series. Of forty tested features, 21 meet the fixed two-partition confirmation rule. The discovery-selected constant is AutoETS. On confirmation series, the fixed selector has mean MASE 3.3501, the constant 3.2570, and the future-informed oracle 3.0005. The selector's error is 0.0931 higher than the constant on this fixed split. This descriptive outcome is not a significance test of selection benefit. The oracle uses future outcomes and cannot be deployed as a selection policy.

Exclusions by reason: zero or unavailable MASE scale: 1.

Confirmation method	Mean MASE
AutoTheta	3.2195
Naive	3.2216
SeasonalNaive	3.2216
Theta	3.2228
SES	3.2263
AutoETS	3.2570
Holt	3.6217
HoltWinters	3.6217

Table 6: All forty prespecified features. D/C denote discovery/confirmation. Signed Kendall effects, BY-adjusted probabilities and the fixed confirmation decision are retained, including negative outcomes. Available counts and raw probabilities are in the public artifact.

Feature	n_D	τ_D	q_D	n_C	τ_C	q_C
acf_1	2946	0.012	1	1280	0.032	0.5
acf_7	2946	-0.004	1	1280	0.017	1
catch24_CO_Embed2_Dist_tau_d_expfit_meandiff*	2946	0.126	4.8e-23	1280	0.154	7.4e-15
catch24_CO_FirstMin_ac	2946	-0.063	3.4e-06	1280	-0.049	0.068
catch24_CO_HistogramAMI_even_2_5	2946	0.006	1	1280	0.004	1
catch24_CO_f1ecac*	2946	-0.094	4e-13	1280	-0.079	0.00023
catch24_CO_trev_1_num*	2946	0.145	4.7e-30	1280	0.173	3.9e-18
catch24_DN_HistogramMode_10	2946	-0.021	0.5	1280	-0.047	0.086
catch24_DN_HistogramMode_5	2946	-0.029	0.12	1280	-0.044	0.12
catch24_DN_Mean	2946	-0.035	0.03	1280	-0.045	0.1
catch24_DN_OutlierInclude_n_001_mdrmd*	2946	-0.042	0.0044	1280	-0.059	0.014
catch24_DN_OutlierInclude_p_001_mdrmd*	2946	0.136	1.6e-26	1280	0.167	2.6e-17
catch24_DN_Spread_Std	2946	-0.034	0.034	1280	-0.044	0.12
catch24_FC_LocalSimple_mean1_tauresrat*	2946	0.117	3.7e-20	1280	0.120	3.5e-09
catch24_FC_LocalSimple_mean3_stderr	2946	-0.003	1	1280	-0.023	1
catch24_IN_AutoMutualInfoStats_40_gaussian_fmmi*	2946	-0.088	1.2e-08	1280	-0.106	1.7e-05
catch24_MD_hrv_classic_pnn40	2946	0.010	1	1280	-0.005	1
catch24_PD_PeriodicityWang_th0_01*	2946	-0.077	3.2e-08	1280	-0.098	9.9e-06
catch24_SB_BinaryStats_diff_longstretch0*	2946	-0.074	2.2e-07	1280	-0.099	1e-05
catch24_SB_BinaryStats_mean_longstretch1*	2946	-0.110	1e-17	1280	-0.098	3.8e-06
catch24_SB_MotifThree_quantile_hh	2946	0.006	1	1280	-0.000	1
catch24_SB_TransitionMatrix_3ac_sumdiagcov*	2946	0.046	0.0031	1280	0.085	0.00017
catch24_SC_FluctAnal_2_dfa_50_1_2_logi_prop_r1*	2946	-0.080	3.6e-09	1280	-0.098	8.1e-06
catch24_SC_FluctAnal_2_rsrangefit_50_1_logi_prop_r1	2946	0.001	1	1280	-0.035	0.44
catch24_SP_Summaries_welch_rect_area_5_1*	2946	-0.064	2.2e-06	1280	-0.054	0.029
catch24_SP_Summaries_welch_rect_centroid*	2946	0.097	1.8e-11	1280	0.076	0.0025
history_length*	2946	-0.149	7.8e-26	1280	-0.156	1.4e-12
kurtosis	2946	0.029	0.11	1280	0.046	0.093
linear_trend_r*	2946	0.114	3.8e-19	1280	0.147	1.4e-13
log_std	2946	-0.035	0.034	1280	-0.044	0.12
lz_binary*	2946	0.041	0.007	1280	0.057	0.02
mse_1	2946	0.002	1	1280	-0.006	1
mse_10	2476	0.020	0.76	1071	-0.017	1
mse_2	2941	0.009	1	1278	0.003	1
mse_5	2834	0.010	1	1223	0.020	1
profile_max	2946	0.018	0.87	1280	0.014	1
profile_median*	2946	-0.050	0.00041	1280	-0.056	0.022
reversal_1*	2946	0.130	2e-24	1280	0.141	1.3e-12
skewness*	2946	0.083	1.6e-10	1280	0.094	8.1e-06
weekly_phase_strength*	2946	0.090	3.3e-12	1280	0.086	5.5e-05

* Rejected at BY $q \leq 0.05$ in both partitions with the same effect direction.

The evaluator retains external per-origin forecast-path audits and independently rescores every exported MASE before compact export. Its resumable cache binds source bytes, fixed protocol, implementation and package identities. Admission independently checks inventory, duplicate separation, temporal schedules, discovery-only imputation, aggregate scores, corrected probabilities and Kendall effects. The public notebook retrains the selector from admitted discovery rows and repeats reference statistical calculations without claiming a new fit of the eight external-corpus forecasting models. These layers distinguish full forecasting reproduction, compact-table statistical reproduction and browser inspection.

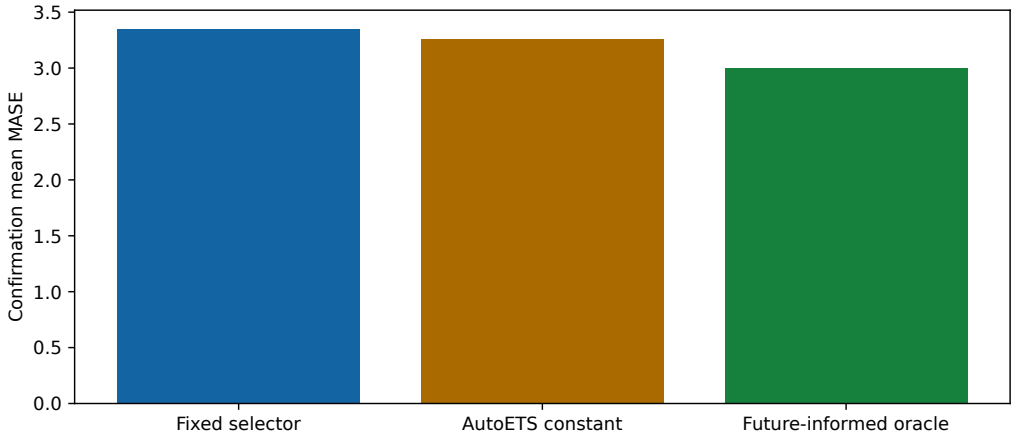


Figure 6: Mean MASE on confirmation series for the fixed discovery-trained selector, the discovery-selected constant method and the future-informed oracle. Bars are descriptive averages over the same admitted confirmation rows; they are not confidence intervals.

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